DATA ITEM DESCRIPTION

Title: EXPLOSIVE ORDNANCE DISPOSAL DATA

Number: DI-SAFT-80931C
AMSC Number: N10331
DTIC Applicable: No
Preparing Activity: OS
Applicable Forms: N/A

Approval Date: 20220711
Limitation: None
GIDEP Applicable: No
Project Number: SAFT-2022-001

Use/relationship: This source data shall contain essential information that is used by the Naval Surface Warfare Center Indian Head Division (NSWCIHD), the U.S. Army Combat Capabilities Development Command (CCDC) Armaments Center Explosive Ordnance Disposal (EOD) Division, and individual Service-specific weapons Research, Development, Test, and Evaluation entities to develop, test, verify, validate, and publish joint service non-nuclear EOD 60-series technical manuals.

This Data Item Description (DID) applies to U.S. and foreign weapons under U.S. acquisition containing energetic material that may be subject to disposal by qualified EOD technicians. This DID applies to weapons programs where a released baseline is being changed or accepted due to an Engineering Change Proposal, waiver, or deviation.

NOTE:

Research, development, test, and evaluation efforts that are not under U.S. acquisition and efforts where testing and evaluation are not under government ownership as part of an acquisition are exempt from this DID (see DoDD 5160.62, DoD Executive Agent for Military Explosive Ordnance Disposal Technology and Training [copies of DoDD 5160.62 are available from https://www.esd.whs.mil/dd]).

NOTE:

This DID applies to any munition category or type, system, or sub-system that contains explosives or energetic materials regardless of specifically identified categories that have been called out in the DID. Examples (not all inclusive) of munition categories or types are mines, missiles, and rockets.

This DID provides the specific requirements for source data from programs and their contractors, using energetic materials within a given system, sub-system, assembly, sub-assembly, or component. This source data is used to generate EOD-specific, 60-series technical manuals, informing qualified EOD technicians on recommended disposal and render safe procedures, mitigating explosive hazards, and allowing for maneuver on the battlefield.

EOD technicians will use this data in support of testing, development, and operational evaluation of EOD procedures on new and modified non-nuclear weapons systems. These weapons systems include manned and unmanned systems (UxS) and launch platforms employed on the ground, in the air, and underwater. This includes platforms and vehicles containing energetic material (i.e., explosive bolts, ordnance, and fuzing), as well as platforms and vehicles carrying ordnance and using non-energetics such as directed energy systems.

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

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NOTE:

Non-energetic platform data is required to ensure safe access and prevent accidental discharge of any directed energy measure or countermeasure while EOD is mitigating explosive hazards.

NOTES:

In accordance with distribution restrictions, EOD data, procedures, and 60-series technical manuals are not releasable outside of the EOD career field; distribution of these documents shall normally be limited to organizations assigned graduates of the U.S. Naval School, Explosive Ordnance Disposal (NAVSCOLEOD) who are actively working in a position that requires access to the information to perform EOD missions. Automated Explosive Ordnance Disposal Publication System (AEODPS) accountholders must be NAVSCOLEOD graduates.

Due to the restrictions, the final published EOD manuals will not be available for delivery to the weapon system program, but the safety board that supports the specific program will be made aware when the EOD procedures are fielded in accordance with DoDD 5160.62 and service-specific requirements.

a. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in the solicitation.

b. This DID will be used by all Department of Defense (DoD) munition acquisition programs when explosive, pyrotechnic items or other hazardous materials that contain energetics are inherent in a system, subsystem, vehicle, aircraft, or munitions being developed, modified, or procured as Non-Developmental Items.

c. This DID supersedes DI-SAFT-80931B.

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format.

2.1. Introduction. The source data package is critical in the building of a joint service-approved EOD technical publication. Therefore, the information provided shall be able to be easily extracted and imported into the EOD publication development system. This will streamline production of that EOD technical publication and also ensure the highest possible quality and technical accuracy.

2.2. Media. Electronic files shall be provided on CD or DVD media.

2.3. Security Markings. Security markings shall be included as follows:

a. Include the applicable distribution statement (i.e., A, B, C, D, E) in accordance with DoDI 5230.24, Distribution Statements on Technical Documents, Enclosure 4 (copies of DoDI 5230.24 are available from https://www.esd.whs.mil/dd).

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b. Include the applicable Controlled Unclassified Information (CUI) markings, as necessary, in accordance with DoDI 5200.48, Controlled Unclassified Information (CUI) (copies of DoDI 5200.48 are available from https://www.esd.whs.mil/dd), and the DoD CUI Registry (the DoD CUI Registry is available online at https://www.dodcui.mil/Home/DoD-CUI-Registry).

c. Include the applicable classification marking, as necessary, in accordance with DoDM 5200.01, Volume 2, DoD Information Security Program: Marking of Information (copies of DoDM 5200.01 are available from https://www.esd.whs.mil/dd).

2.4. Electronic Files. In all cases, the original file format shall be provided with the highest resolution possible. Screen or low resolution of text or visual information will result in rework and poor quality.

2.5. Copyright and Proprietary Information.

a. Technical manuals shall not contain copyrighted material except as specified in the Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) (copies of the FAR and DFARS are available from https://acquisition.gov/). When copyrighted material is to be included in a technical publication, the developer shall obtain prior written permission from the copyright owner or authorized agent for its use. The signed, written permission shall be delivered together with the final reproducible copy. The written permission shall contain a statement declaring whether or not a copyright credit line is required. When it is necessary to include copyright and proprietary material, it shall be clearly identified and the following warning statement shall be included on the title page:

"This document contains copyright or proprietary materials. Infringement of copyright or proprietary material may violate existing Federal laws and statutes and result in criminal penalties, imprisonment, or removal from office."

b. Label all copyrighted and priority information clearly so when it is extracted and placed into the EOD publication, it can be properly labeled as such.

2.6. Textual Information.

a. File formats: Acceptable file formats for text shall be Microsoft Word (.DOCX file extension), Microsoft PowerPoint (.PPTX file extension), or Adobe Portable Document Format (.PDF file extension).

b. Security controls: Files should not be locked, password-protected, or restricted. Printing and copying content shall be allowed.

2.7. Visual Information.

a. Size and resolution: Visual information shall fit onto a single 8.5- x 11-inch (letter size) sheet of paper when printed. Visual information smaller than 4 x 6 inches shall not be provided. Photographs and raster images should be provided at a minimum of 300 dots per inch.

b. Photographs shall be provided as follows:

(1) Provide high resolution originals of any photographs.

(2) If possible, remove any background or photograph items on a white background.

(3) If using embedded content (e.g., a Microsoft Word file with an embedded photograph), provide the photograph as a separate, standalone image file.

(4) Preferred file formats are JPEG or TIFF (with LZW compression).

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c. Callouts.

(1) Callouts should be clear and legible.

(2) Any original files used to embed callouts (i.e., Microsoft Word, Microsoft PowerPoint, Adobe Photoshop, or Adobe Illustrator) shall be provided.

(3) The original files without callouts (see figure 1) shall be provided. This will be useful if callouts have to be changed or removed. Arrows should also be removed in the version without callouts.

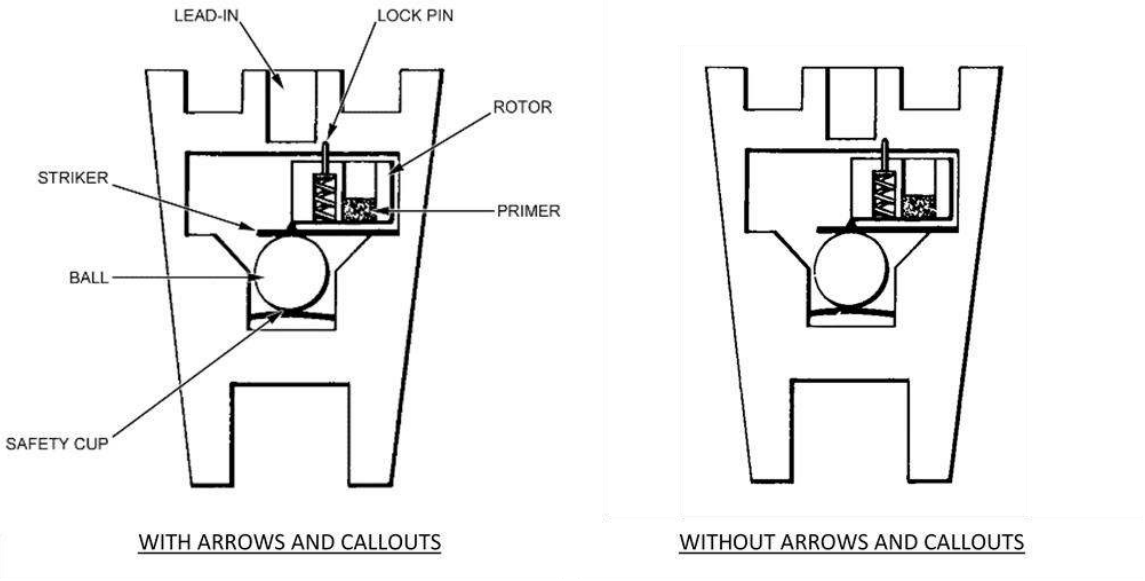


FIGURE 1. Visual information with and without arrows and callouts.

d. 3D models and computer aided design (CAD).

(1) If available, original source files shall be provided (e.g., AutoCAD or SolidWorks).

(2) Because of the multitude of CAD systems and associated file formats (e.g., CATIA, SolidWorks, NX, Creo, and Inventor), 3D PDF shall be a viable alternative solution to providing the original files. Original file formats are typically very large and stored in a proprietary format and require the native software to open and view the content.

(3) A 3D PDF is a PDF that is small in file size and can house CAD data, its metadata, and other files (see figure 2 for example 3D PDFs). A 3D PDF also has security features and may have the ability to demonstrate model animations. In addition, this file format is easily imported into an EOD publication and will facilitate knowledge transfer and improve comprehension.

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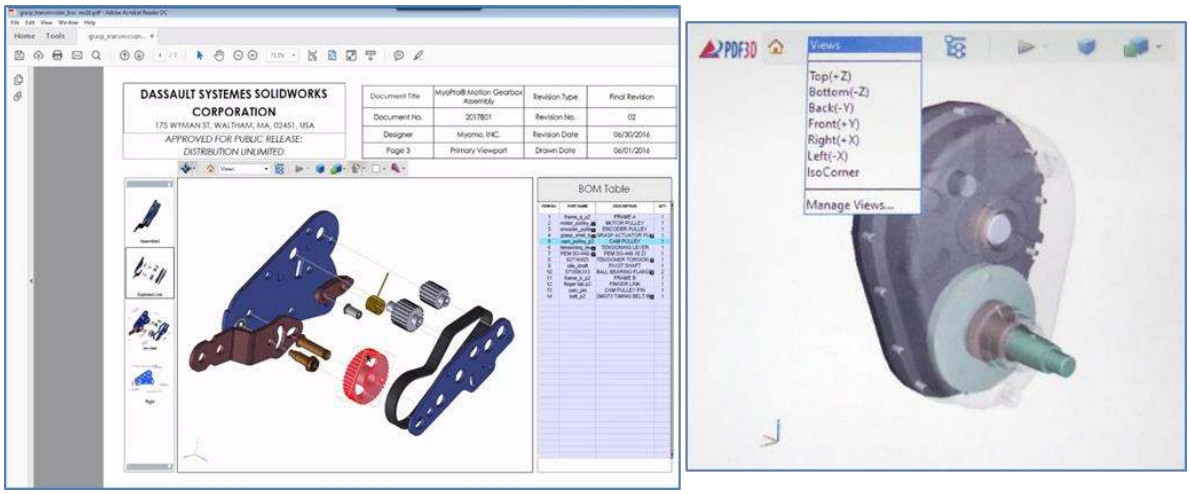


FIGURE 2. Examples of 3D PDFs.

e. Video. Video data shall be provided as follows:

(1) Provide any original files used (e.g., Adobe Premiere or Apple Final Cut).

(2) Preferred file format is MPEG-4 (.MP4 file extension).

(3) Alternative file formats are MPEG (.MPG file extension), QuickTime Movie (.MOV file extension), or Windows Media Video (.WMV file extension).

3. Content.

3.1. Data Requirements for All Explosive and Energetic Items. The EOD data shall contain an official cover sheet that lists the office, vendor, or manufacturer drafting the data; author name and contact information; date published; contract number, revision sequence, and supersession dates; and distribution statement and applicable security classification markings. Data shall be as follows and shall be formatted in the following main and sub-section format:

a. Safety summary: Add any additional safety precautions. These are general precautions that do not appear in the body of the information set, but are applicable either throughout the information set or to a broad area of it. The following examples contain formatting and content parameters for warnings, cautions, and notes.

WARNING

An operating or maintenance procedure, practice, condition, or statement, which, if not strictly observed, could result in injury to, or death of, personnel or long-term health hazards.

CAUTION

An essential operating or maintenance procedure, practice, condition, or statement, which, if not strictly observed, could result in damage to, or destruction of, equipment or loss of mission effectiveness.

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NOTE:

Any other pertinent procedural, system specific, or other information that does not fall under a warning/caution, but is pertinent to the system.

b. Identification: munition type, size, type by function, nomenclature.

c. Designation: the official designation or nomenclature of the ordnance, any common names, and any known foreign designations. Include National Stock Number, manufacturer part number, and applicable Department of Defense Identification Code (DODIC).

d. Type: a detailed explanation of the type by function including any fuzing, stabilization, guidance, and launch platforms.

e. Use: what the weapon or munition is used for including any optional applications.

f. Paintings and markings: all external markings, paintings, and colors that will appear on the end item. Typical markings include ordnance designation, loading information, settings, index marks, time scales, instructional notes, lot numbers, color code designators, and manufacturer's markings and symbols.

g. External features.

(1) All fittings and features that will differentiate the item from a similar item within a class or family of ordnance items. Features or fittings should be visible and obvious without touching or disassembling the item.

(2) Line drawings with appropriate callouts identifying the pertinent fittings, features, and dimensions of the item.

h. Ordnance used with: provide a list of all associated systems this item is used with.

(1) Fuze data shall state what bombs, dispensers, projectiles, landmines, grenades, or rocket warheads it is used in.

(2) In case of bombs, projectiles, or rocket warheads, etc., state what fuzes, rockets, fin assemblies, or guidance units are used.

(3) Dispensers, clusters, launchers, and certain projectiles shall include all submunitions that could be loaded.

(4) Underwater ordnance items such as torpedoes, naval mines, depth charges, sound signals, and markers shall include data on what exploders, extenders, clock starters, hydrostatic arming devices, or target detection units can be used.

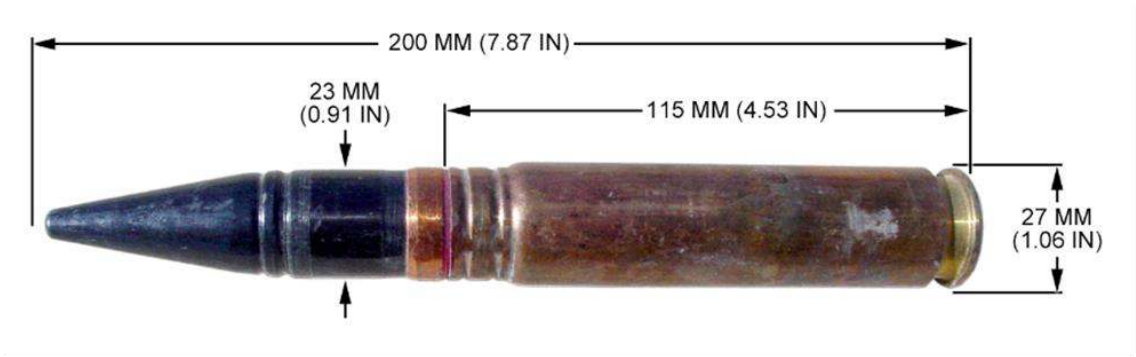
(5) Delivery system utilized (e.g., vehicle or platform nomenclature and specific launcher).

i. Visual information.

(1) Provide key measurements in visual context (i.e., length, exposed length, width, height, and diameter).

(2) Provide visual information files both with and without callouts and arrows (see figure 3).

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WITH ARROWS AND CALLOUTS



WITHOUT ARROWS AND CALLOUTS

FIGURE 3. Example of visual information with dimensional information.

(3) Provide visual information of key identification features (see figure 4) and any critical components or subcomponents. Examples include fuze well, set screw, lug, bourrelet, rotating band, base plug, shipping plug/cap, spring, fin, firing pin, shroud, cover, spacer, base plate, and wrench groove. Include any specific markings, symbols, color bands, stencils, lot numbers, or paintings.



FIGURE 4. Visual information of key identification features.

(4) Provide visual information with multiple and numerous camera angles of the round or item. Include top, bottom, left, right, front, back, nose, and base views of the complete item (see figure 5).

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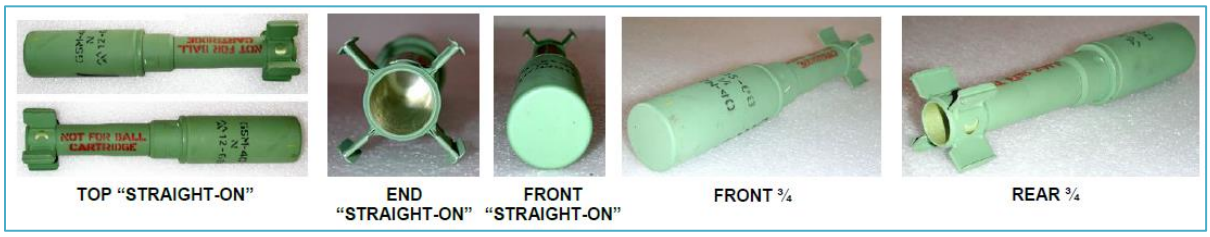


FIGURE 5. Visual information from multiple viewpoints.

(5) Provide identification and description visual information as shown in figure 6.

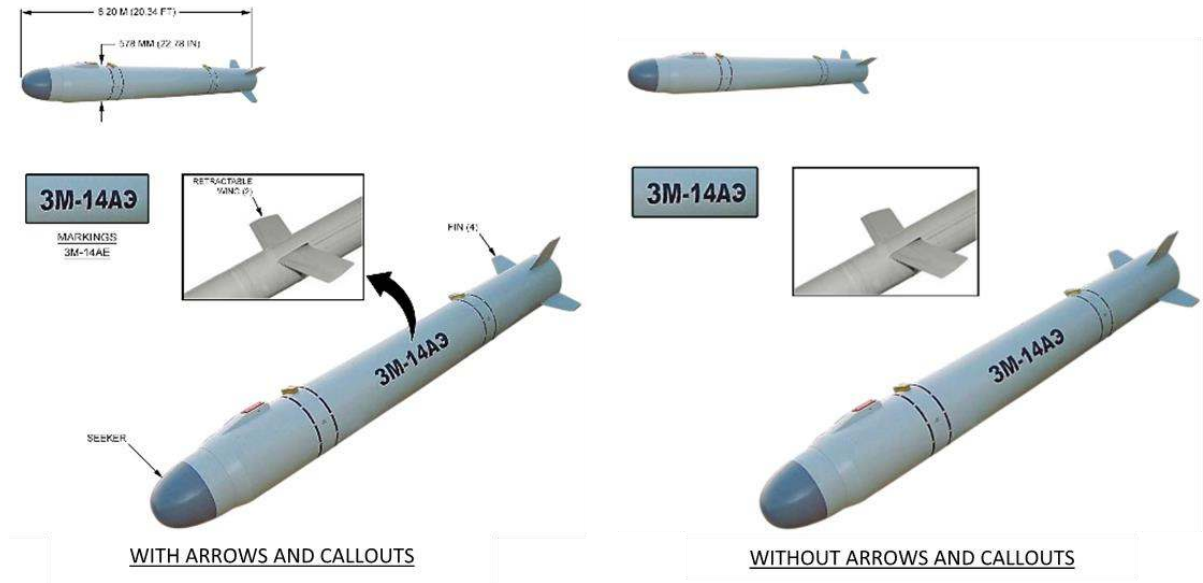


FIGURE 6. Visual information to support identification and description section.

3.2. Description. The description shall include the following:

a. Describe major sections, subassemblies, classified materials, and hazardous components or materials of the system and their purpose, function, and method of operation. This shall include figures, pictures, and illustrations of all components with callouts of external views and internal cutaways specifically noting major sections, subassemblies, hazardous components, and power sources, as well as arming, safing, and firing subcomponents. An overall narrative physical description and arrangement of the item to include item material, weight, and dimensions in both inch-pound and metric shall be included; this shall include the following:

(1) Overall length

(2) Diameter, minimum and maximum

(3) Propellant volume

(4) Empty weight

(5) Filler volume

(6) Fuze exposed length, minimum and maximum

(7) Height, minimum and maximum

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(8) Width, minimum and maximum

(9) Oxidizer volume

(10) Oxidizer weight

(11) Propellant weight

(12) Filler weight

(13) Thickness, minimum and maximum

(14) Total weight

(15) Total length, minimum and maximum

b. Describe the appearance and condition of the item in the unarmed, pre-armed, and armed conditions.

(1) Composition and shell material. Include material specifications and structural dimensions such as type of material (i.e., steel, aluminum, and bakelite), thickness of material, and thread specifications.

(2) Describe all toxic or hazardous materials, liquids, explosives, hydraulics, or pneumatic pressure sources and routing, fuels, and oxidizers.

(3) Describe electric voltage sources and circuits, thermal or chemical energy sources, and mechanical hazards. Refer to section 3.10 for more information.

(4) Describe all explosive main-charges, subassemblies, and firing trains. Include type; weight (metric and inch-pound), both individually and net explosive weight (NEW); and unique sensitivities and composition, including specific ingredients and their percentages.

(5) Describe any other item that could present a physical threat to the safety of personnel (e.g., spring-loaded guidance fins and power-actuated access panels). Include any known Environmental Protection Agency-approved disposal information relating to the above items.

(6) Describe any design elements intended for Insensitive Munition (IM) compliance and where they are located. Also provide data regarding IM verification testing as any failures in IM compliance testing can provide insight to possible disposal methods.

3.3. Hazardous Components. Hazardous component data shall include the following:

a. List of all hazardous components, toxic materials and liquids, explosives, hydraulic or pneumatic pressure sources, mechanical hazards, high voltage sources and circuits, thermal or chemical electrical energy sources, fuels, oxidizers, and other items that could present a physical threat to the safety of personnel. Include location, name of component, quantity, and weight (metric and inch-pound). Also, provide individual component part numbers as applicable. Table I is an example of the format for listing hazardous components.

(1) Additionally, for aircraft and UxS, UxS general information. Include general information on major systems including toxic or hazardous materials or liquids to include type, weight, and location. Include information on pressurized vessels, flares, markers, fire suppression, pneumatic, hydraulic, electrical auxiliary power units, and hazard zones in, on, and around UxS. Items shall be described in sufficient detail to provide positive identification and location with graphic illustration callouts (e.g., diagrams, photographs, computer graphic illustrations, and line drawings).

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TABLE I. Example of detailed list of hazardous components.

Location	Part Number	Item	Material	Parts by Weight	
				Metric	Inch-Pound
Warhead Assembly: ESAD	DXXX-XX	RSI-390: Initiator, Plastic Encapsulated, Snap In	RSI-007 (CL-20)	0.38 g	0.0134 oz
	DXX-X	Booster	PBXN-5	4.54 g	0.16 oz
	DXXX-X	High Explosive Charge	PBXN-9	1.084 kg	2.39 lbs
Electronics Sled Assembly	Eagle Picher XXXX-X	Thermal Lithium Battery	Boron calcium chromate and zirconium ferric oxide	0.10 g	0.0035 oz
Rocket Motor Assembly: CDU	DXXX-XX	RSI-369: Exploding Foil Deflagrating Initiator with Integrated Barrier (EFDI with IB)	RSI-007 (CL-20) Input Pellet	0.0055 g	0.000194 oz
		RSI-369: EFDI with IB	BKN03 Output Pellet	0.238 g	0.0084 oz
		BKNO3 Ignitor Bag	BKN03	3.05 g	0.1076 oz
		Sleeve and Propellant – Ring Motor	Blue Thunder (75 to 82% ammonium perchlorate, 1 to 5% aluminum)	532.5 g	18.783 oz
Bladder and Antenna Assembly	DXXX-XX	Fuel Bladder with Liquid Fuel	JP-10 (2.08 L/0.55 gal)	N/A	
Total NEW				1.62 kg	3.57 lbs
NOTE: Abbreviations and acronyms used in the above table include Electronic Safe and Arm Device (ESAD), grams (g), kilograms (kg), ounces (oz), liters (L), gallons (gal), and pounds (lbs).					

3.4. Operational Sequence and Component Functioning. The data shall include the following:

a. Step-by-step operation system from initial receipt of power, including firing sequence and arming sequence.

(1) Describe the normal operation sequence of the ordnance item step-by-step to include design safety features, environmental sensing devices and their operating parameters, pre-arming, arming, and functioning to include environmental physical situations required to function device once arming is complete (for example, g-force required to overcome an anti-creep spring holding back a firing pin or the change in magnetic gradient to activate a mine).

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(2) Visual information: Illustrations, photographs, drawings, and associated callouts will be provided and will support the functioning descriptions. Visual information will include callouts for all components mentioned in the sequence of operations or functioning. Cutaway drawings or radiographs will be provided to illustrate internal working. Electrical block and circuit diagrams of arming/firing system will be provided for electrically initiated ordnance items and will include all firing capacitor switches, bleeder resistors, and power sources. Exploded view illustrations may aid with component functioning as shown below. Provide visual information files both with and without callouts and arrows (see figures 7, 8, and 9 for examples with callouts).

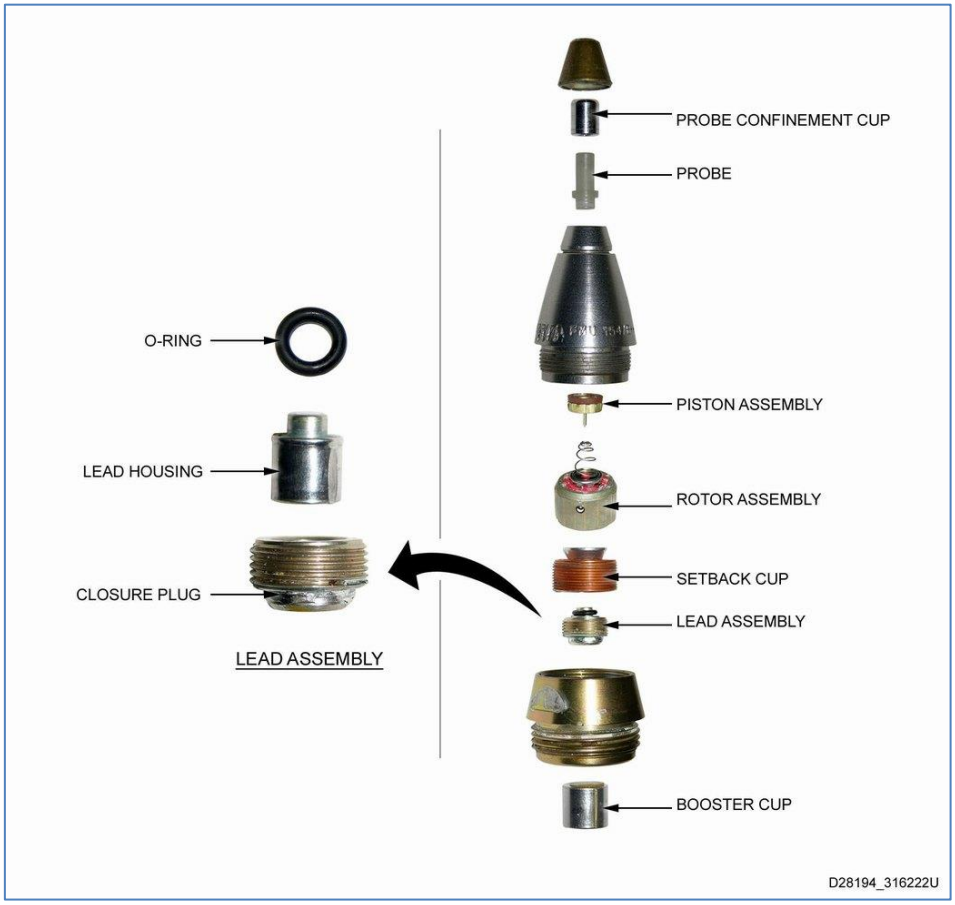


FIGURE 7. Example visual information (exploded view) to support functioning section.

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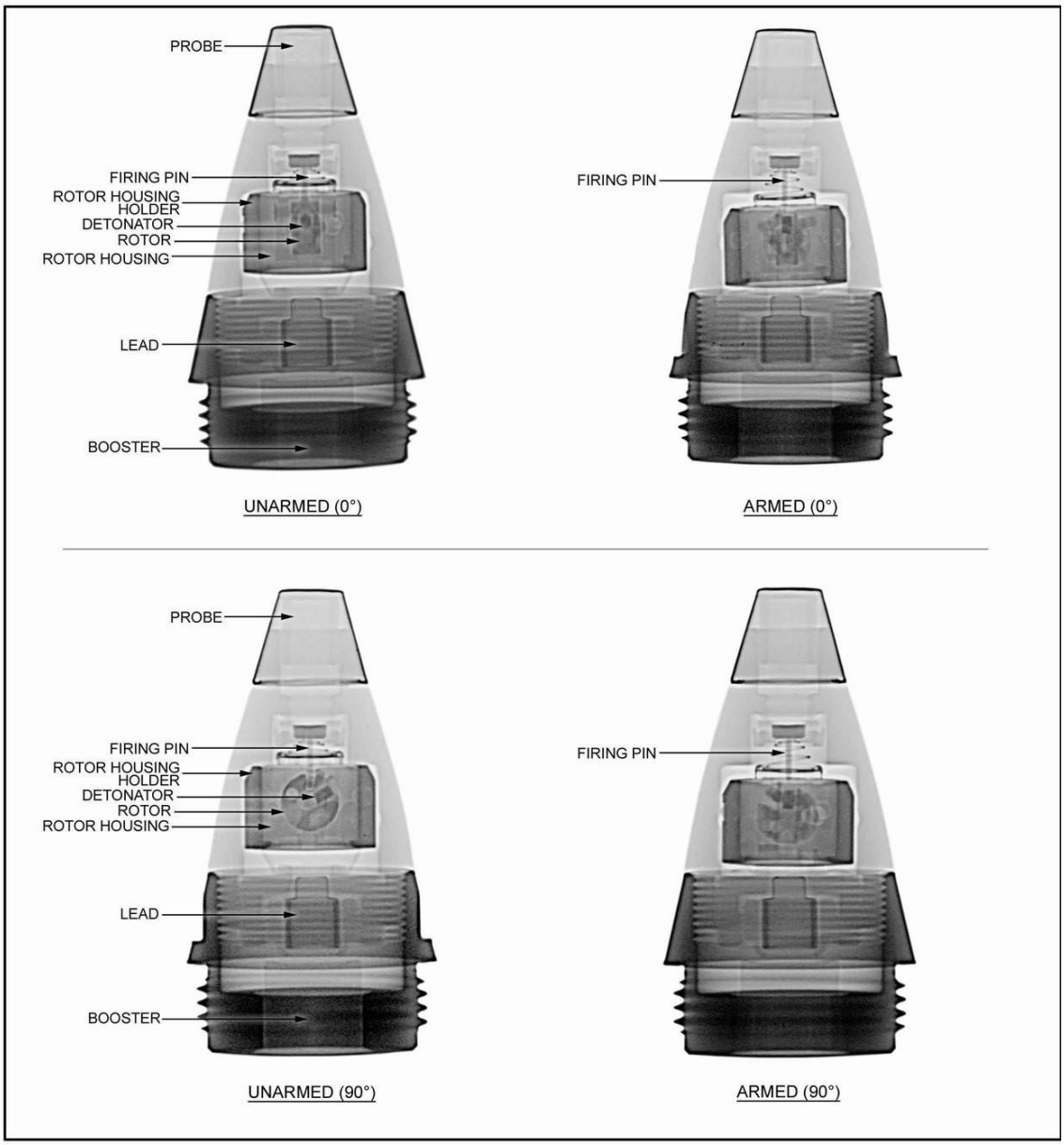
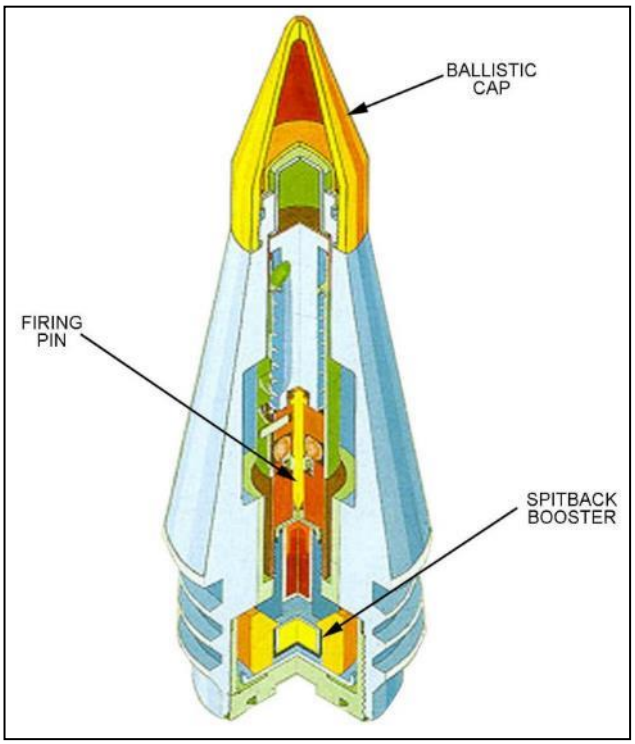


FIGURE 8. Example visual information (radiograph) to support functioning section.

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CUTAWAY ILLUSTRATION

FIGURE 9. Example visual information (cutaway illustration) to support functioning section.

3.5. Appearance. If there are no external means to distinguish between armed and unarmed conditions, the parameters that would make the item armed or unarmed shall be described as follows:

a. Unarmed condition: Describe the physical appearance and condition of the item in the unarmed condition. Describe the item so it can be positively identified as unarmed by the physical condition or appearance.

b. Armed condition: Describe the physical appearance and condition of the item in the armed condition. Describe the item so it can be positively identified as armed by the physical condition or appearance.

c. Other means of determining safe and armed conditions: Describe any additional methods to aid in determination of armed versus unarmed condition (i.e., data logs, electronic or remote query).

d. Visual information: Provide visual information that shows the item in various conditions or states (i.e., with and without safety pins, armed and unarmed conditions, with and without fuze, and extended and closed fins).

3.6. Recommended EOD Procedures. Recommended render safe procedures for armed and unarmed conditions of the item, recommended disposal procedures as provided by the respective vendor or manufacturer, and recommended download procedures shall be as follows:

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NOTE:

Safety warnings, cautions, and notes shall be incorporated into the recommended procedures, where applicable.

NOTE:

Each page containing recommended render safe procedures shall be stamped with the following: "RECOMMENDED RENDER SAFE PROCEDURES."

NOTE:

Indicate what procedures have and have not been tested as part of the program. Additionally, indicate the conditions tested and validated, if any (e.g., armed and unarmed).

SPECIAL WARNING

Indicate procedures are untested and are based on the best technical data available.

a. Unarmed procedures.

(1) Render Safe Procedure: describe, sequentially, the steps to render the item safe. This includes installation of safing or protective devices (or both), disassembly, disconnections, tools required, recovery actions, transportation, and handling. Include computer graphics or drawings as appropriate. Also include separate procedures for varying item configuration, if applicable, for each condition listed under appearance. Describe the intended objective of the procedures (or intent) and criteria to ascertain whether the procedure was successful.

b. Armed procedures.

(1) Render Safe Procedure: describe, sequentially, the steps to render the item safe. This includes installation of safing or protective devices (or both), disassembly, disconnections, tools required, recovery actions, transportation, and handling. Include computer graphics or drawings as appropriate. Also, include separate procedures for varying item configuration, if applicable, for each condition listed under appearance. Describe the intended objective of the procedures (or intent) and criteria to ascertain whether the procedure was successful. Procedures should be developed in order to be conducted by remote means, if possible.

c. Disposal procedures.

(1) Describe explosive ordnance disposal procedures (normally detonation or burning) for hazardous materials for both armed and unarmed conditions. Include procedures for the complete item as well as all individual hazardous and classified components.

(2) Include computer graphic illustrations, drawings, or video animation (i.e., fly-throughs) as appropriate. The objective of disposal procedures is to ensure all energetic materials and classified components are consumed by the procedures. NOTE: Disposal procedures are considered successful when consumption of energetics is commensurate with consumption of a fully functioning All-Up-Round.

d. Disassembly procedures.

(1) If disassembly is possible, describe step-by-step the normal disassembly of the item required to separate the explosive and hazardous components from the item body. This includes installation of safing devices, protective devices, cable disconnections, and proprietary or specialized tools required. Include computer graphics illustrations or drawings as appropriate to help illustrate difficult procedures. Include separate procedures for varying item configuration if possible.

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3.7. Supplemental Information. The data shall include the following supplemental information, as applicable:

a. Known or suspected information on sensitivity of weapons to electrical, magnetic, electromagnetic, chemical, explosive, heat, radiological, mechanical, or other energy that could activate or function item if armed, unarmed, damaged, or undamaged.

b. Blast sensitivity, fragmentation range, and size of fragments of the main charge or warhead.

c. Test report or other documents detailing design safety features and operational and safety tests.

d. Safety protection requirements for hazardous and toxic solids, liquids, or gaseous materials likely to be encountered.

3.8. Download Procedure. If applicable, the data shall include the download procedure as follows:

a. Describe, sequentially, the steps required to download and separate the item from the platform, regardless of manned or unmanned, launcher, mount, or canister. This includes access, installation of safing or protective devices (or both), disassembly, disconnections, tools required, recovery actions, transportation, and handling. Include computer graphics, drawings, or video animation as appropriate. Also, include separate procedures for varying item configuration, if applicable, for each condition listed under appearance.

b. Include step-by-step non-nuclear munitions access and downloading procedures. Include removal of racks, dispensers, launchers, and gun systems. Include actual photographic pictures (if available). Diagrams, computer graphic illustrations, line drawings with callouts of components referenced in the description(s), or video animation are also acceptable.

3.9. General Platform Data. General platform data is data relative to the launch platform: ground, airborne, submersible, and UxSs. This data shall include the following:

a. EOD requires unique written vehicle access procedures to support the safe and efficient recovery and disposal of battlefield ordnance remains during Battle Damage Assessment and Repairs.

b. EOD must be able to enter a combat vehicle after it has been damaged in order to safely remove explosive ordnance.

c. Towed artillery that does not have onboard storage, does not require power-down procedures, and is covered by general stuck-round procedures is excluded from this requirement.

d. A list of non-nuclear munitions compatible with or used with a given platform.

e. A brief description of the type of platform.

f. All exterior painting and markings.

g. Platform dimensions including front, rear, side, and top views (e.g., diagrams, photographs, computer graphic illustrations, and line drawings).

h. Platforms with onboard ordnance storage in a lockable compartment (e.g., a main battle tank or infantry-fighting vehicle), guided missile launchers, and troop transport vehicles with a lockable compartment (e.g., armored personnel carriers or Mine Resistant Ambush Platforms) shall provide any safety procedures related to lockouts, allowing for safe personnel access without inadvertent actuation.

i. Any platforms and systems with integral explosive components (e.g., explosively activated ejection seats or cabins [i.e., egress systems], explosively formed barrier shields, integral reactive armor, cutters, disrupters, and their components).

(1) Include all explosive components, their location, explosive weight, and type. Items shall be described in sufficient detail to provide positive identification. Include a step-by-step sequential

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operation of the system to include the various initiation methods and functioning times. Include diagrams, photographs, computer graphic illustrations, and line drawings with callouts of all components covered in the sequence of operations.

j. Any other safing procedures for mechanical hazards.

k. Methods to isolate all platform electrical power sources (e.g., removal of platform batteries).

l. Directed energy and other energetic devices not containing explosive materials:

(1) Describe system to include access, mechanical, and electrical safing procedures, operational sequence, and emergency disassembly. Include diagrams, photographs, computer graphic illustrations, and line drawings with callouts to support descriptions. Provide nonionizing radiation protection studies that include nominal ocular hazard distances, nominal skin hazard distance, laser safety information, optical density requirements for eye protection, output parameters of the lasers, and hazardous optical laser radiation exclusion areas.

m. Data specific to unmanned aerial vehicles (UAV) in addition to general platform data:

(1) Describe in sequential detail the UAV prelaunch, launch, post-launch, flight, arming, and firing functionality of the system. Provide specifics on any arming and firing parameters or environments that affect the system functionality (e.g., after the UAV departs the launcher, the umbilical harness is disconnected between the UAV and the launcher, allowing arm power to the ESAD).

n. Data specific to manned aircraft, in addition to general platform data:

(1) Cockpit entry, both normal and emergency.

(2) Engine shutdown procedures.

(3) Landing gear system and safing procedures.

(4) Survivability life support systems. Identify, describe, and provide diagrams, photographs, computer graphic illustrations, line drawings, etc. with callouts of all components that have a potential hazard to personnel (such as pressurized oxygen bottles, flares, markers, and fire suppression systems).

(5) Armament systems. Describe weapons release systems to include racks, launchers, pylons, and flare (or chaff) dispensers. Include functioning sequence, system location, location of explosive components or actuators, mechanical and electrical safing procedures, and explosive component removal procedures. If any component is an already established military item without special modifications, reference to item and location illustrations on aircraft is sufficient (e.g., “MAU-12A ejection racks located at stations 2 and 6”). Include diagrams, photographs, computer graphic illustrations, and line drawings.

(6) Internal gun systems. Describe gun systems to include access, mechanical and electrical safing procedures, operational sequence, and emergency disassembly. Include diagrams, photographs, computer graphic illustrations, and line drawings, with and without callouts to support descriptions.

(7) Include information on any component that presents a hazard to personnel.

3.10. EOD Data Requirements for Weapon System Firing Energy Discharge Testing.

a. System. System data shall include the following:

(1) Official nomenclature (MK/MOD).

(2) NAVSUP Ammunition Logistics Center (NALC)/DODIC.

(3) Ordnance type.

b. Battery information. Battery information data shall include the following:

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NOTE:

Complete for all batteries that supply power to sensors/triggers and Electronic Safe and Arm Fuze (ESAF)/ESAD devices for initiation of energetic components.

(1) Replacement for: (previous battery manufacturer/system part number or not applicable).

(2) Primary manufacturer.

(3) Primary manufacturer part number.

(4) Additional source of supply: (YES/NO).

(5) Additional manufacturer.

(6) Additional manufacturer part number.

(7) Weapon system battery part number.

(8) Used in other ordnance systems: (YES/NO).

(a) If "YES," list ordnance system nomenclature/NALC/DODIC.

(9) Type: (thermal, spin-dependent liquid reserve, or non-reserve).

(10) Chemistry: (anode/cathode).

(11) Quantity used in ordnance system.

(12) Ignition type: (electric, inertia, or no activation required).

(13) Ignition requirement: (all-fire current/duration, g's/duration, etc.)

(14) Ignitor resistance with tolerance.

(15) Maximum output voltage.

(16) Maximum case temperature.

(17) Terminal configuration (provide pinout diagram).

(18) Dimensions.

(19) Weight.

(20) Safe to conduct open circuit test: (YES/NO).

(a) If "NO," minimum load required for safe discharge: (prevent thermal runaway).

(21) Estimated open circuit discharge time to battery voltage less than detonator no-fire voltage threshold. Or, if a high-voltage detonator is used, battery voltage less than voltage required to maintain a sufficient charge on the firing capacitor.

(22) DoD Hazard Class (battery only).

c. Electronic safe, arm, and fire system/device. The system data shall include the following:

(1) Provide schematic showing path from power source to firing capacitor and firing capacitor energy dissipation resistor network (bleed path). Include firing capacitor and bleeder resistor values and tolerances.

(2) Minimum battery operational voltage threshold to maintain a charge on the firing capacitor(s).

(3) Minimum battery operational voltage threshold to operate sensor/trigger (e.g., proximity, impact, and time delay).

(4) Detonator/initiator characteristics. Include type (e.g., hotwire or LEEFI):

(a) For high voltage types, provide maximum no-fire voltage.

(b) For low voltage hotwire types, provide the following:

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1. Constant current no-fire value.
2. If applicable, capacitor discharge no-fire value (include capacitance value).
3. Bridge wire resistance value (include tolerance).

(5). Provide arming (charging), firing, and dud firing circuit discharge description.

d. Hardware data requirements. Hardware data requirements shall be as follows:

(1) Batteries:

NOTES:

If an organization other than NSWCIHD has conducted open circuit testing but the quantity tested was less than 10, the test data will be accepted if the affiliated organization provides the test plan and test report. If fewer than 10 batteries of each type have been tested, further data will be required; however, the number of assets for additional data will also be reduced, commensurate with the data requirement for a total of 10 test assets.

Previous open circuit testing must show the battery discharge was monitored to voltage below the detonator no-fire threshold or, for systems using high-voltage detonators, to voltage below the voltage required to maintain sufficient charge on the firing capacitor. Documentation showing open circuit testing to a percentage of peak shall not be accepted if that percentage is greater than the no-fire threshold of the energetic devices.

(a) Pinout diagram and, if applicable, mating connector part number.

(b) Safety data sheet.

(c) DoD hazard class documentation.

(2) Test plan and report data:

NOTES:

For systems with hotwire low voltage type electro-explosive devices, the battery should be monitored from activation until the voltage output falls below detonator/ignitor no-fire threshold. For thermal batteries, case temperature will be monitored to below 49 °C.

If high-voltage detonators (i.e., LEEFI, EBW) are used, batteries will be monitored to below the voltage required to maintain a sufficient charge on the firing capacitor used to initiate the detonator. Batteries that supply sensors only will be monitored until voltage output falls below the sensor's operational voltage threshold.

(a) Test report with voltage and temperature (if thermal) versus time plots for each battery. For thermal or spin-dependent reserve batteries, include pre- and post-activation photos of each test article with the activation indicator and part or serial number clearly visible. For non-reserve types, include a photo of a single battery with nomenclature clearly visible.

(b) If testing is planned (i.e., not already conducted), representatives from NSWCIHD Ordnance Electronics Exploitation Laboratory (Ord-EEL Code D14) and the Military Technical Acceptance Board will review and approve the proposed test plan and attend the Qualification Test during the open circuit portion in order to observe and verify that portion of the test.

3.11. Data Required for Modeling and Simulation. Data shall be provided that includes the following:

a. Detailed geometry of the munition and any launch container associated with the item.

b. Materials information (i.e., a bottom-level drawing package).

c. Information on the reactivity of the energetic components that may exist.

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